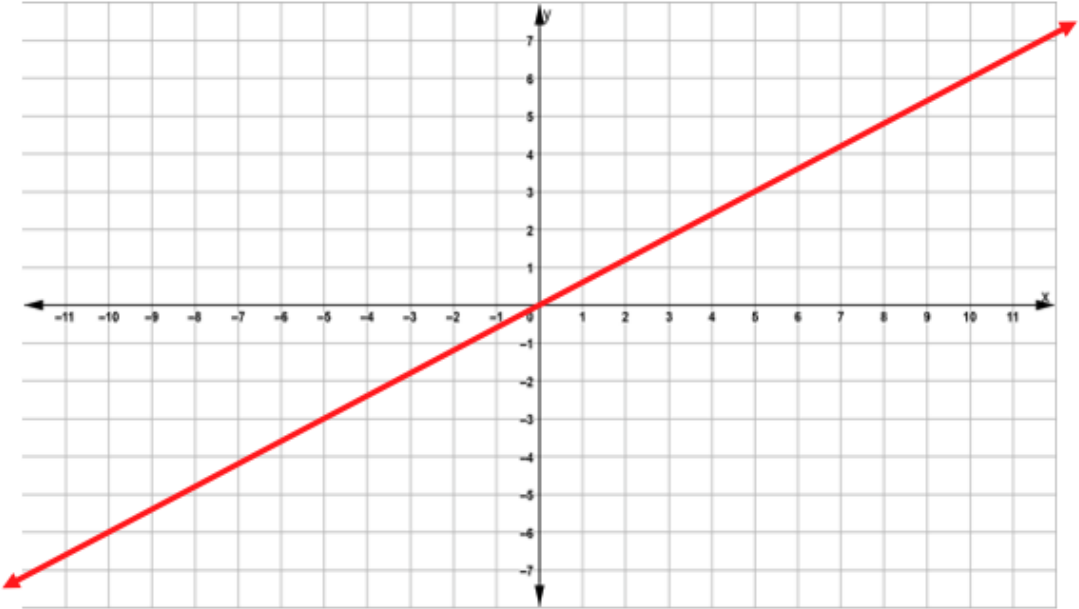


7th Grade Unit 11 Assessment Guide

General Information	Unit 11 continues building students' understanding of proportional reasoning by extending it to multiple algebraic representations. Students also revisit the relationship between the diameter of a circle and the circumference of a circle. This is also the culminating unit in the course for developing an understanding of and applying proportional relationships in two variables. Throughout the course students have been using proportional reasoning, so to end their journey with algebraic representations helps students take their first steps into algebra. If students struggle with one or more of the representations, teachers can utilize contexts from other units to further scaffold student learning. For example, currency conversions can be given, and the student can build from their understanding to create an equation, table, or graph. If a teacher notices that students struggle with fluency and equivalence of rational numbers, teachers are encouraged to consider how to support these students throughout Unit 12 in order to meet the expectation of fluency with operations of rational numbers by the end of the course. Consider not giving students partial credit if they incorrectly write the constant of proportionality as its multiplicative inverse. This is a common misconception and indicates that the student is not understanding the benchmark. At the beginning of the assessment is a box with the statements: <i>Because you are using a calculator, it is possible to accidentally punch the wrong button. To ensure you earn partial credit, show the operation that you entered into the calculator and the result. It is helpful if you label the operation with information from the word problem.</i> This suggestion can also help a student self-scaffold a problem. As you consider the point distribution, consider the sub-steps students may take. When you set up the point distribution consider that not every student needs to show sub-steps so your point distribution should not punish students who can think through the problem and find an answer using a calculator such as the Desmos scientific calculator. It is possible that students can enter these problems as one expression into a calculator such as the Desmos scientific calculator to get the correct answer.
Calculator Information	The questions on this assessment are calculator.
Total Recommended Points	The total recommended points in this guide is 100 points.

Question	1	Benchmark(s)	MA.7.AR.4.2
$\frac{4}{5}$ or 0.8		Recommended Scoring (8 Total Points)	Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors.
Question	2	Benchmark(s)	MA.7.AR.4.3
		Recommended Scoring (7 Total Points)	Students who graph the relationship as $y = \frac{5}{3}x$ are not showing an understanding of the benchmark and should not receive any credit.

Question	3	Benchmark(s)	MA.7.AR.4.4
$x = 0.75y$ or $x = \frac{3}{4}y$ or $y = \frac{4}{3}x$		Recommended Scoring (8 Total Points)	Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors.
Question	4a	Benchmark(s)	MA.7.AR.4.2
$\frac{1}{12}$		Recommended Scoring (6 Total Points)	Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors.
Question	4b	Benchmark(s)	MA.7.AR.4.4
$x = 12y$ or $y = \frac{1}{12}x$		Recommended Scoring (6 Total Points)	If a student makes a minor error in question 4a but correctly writes the equation based on that error, then the student should not lose points for this question.
Question	4c	Benchmark(s)	MA.7.AR.4.3
D		Recommended Scoring (6 Total Points)	Consider not giving partial credit.

Question	5a	Benchmark(s)	MA.7.AR.4.4										
<table><tr><td><i>x</i></td><td>0</td><td>1</td><td>2</td><td>5</td></tr><tr><td><i>y</i></td><td>0</td><td>1.4</td><td>2.8</td><td>7</td></tr></table>		<i>x</i>	0	1	2	5	<i>y</i>	0	1.4	2.8	7	Recommended Scoring (15 Total Points)	Consider giving students 5 points for each missing value in the table. Students can use the graph to read the value for $x = 1$. Students will have to determine the proportional relationship to complete the other two missing values.
<i>x</i>	0	1	2	5									
<i>y</i>	0	1.4	2.8	7									
Question	5b	Benchmark(s)	MA.7.AR.4.4										
$y = 0.3x$		Recommended Scoring (8 Total Points)	Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors.										
Question	6	Benchmark(s)	MA.7.AR.4.1										
☐ Santa Rosa Island ☐ Tiger Creek Preserve ☐ Jonathan Dickinson State Park ☐ Kissimmee Prairie Preserve State Park ☐ Lake Wales Ridge National Wildlife Refuge		Recommended Scoring (18 Total Points)	Consider giving students 6 points for each correct choice and deducting 6 points for each incorrect choice. Students who select all choices should receive 0 points.										

Question	7	Benchmark(s)	MA.7.GR.1.3
The graph shows that the relationship between the circumference and the diameter is ☒ proportional. ☐ not proportional. The constant of proportionality ☐ is 2. ☐ is 3. ☒ is π. The equation that represents the proportional relationship is ☐ $y = 2x$. ☐ $y = \frac{3}{x}$. ☒ $y = \pi x$. ☐ $y = \pi x^2$.		Recommended Scoring (18 Total Points)	Consider giving students 6 points for each correct answer box.